



The Ekpyrotic universe

The Ekpyrotic or cyclic universe paradigm, an alternative to the standard big bang inflationary model <https://digit.in/oct19-01>



Hawking points

The paper by Penrose that claims to present observational evidence of Hawking points in the CMB <https://digit.in/oct19-02>

OLDER THAN THE UNIVERSE

Can there be relics from before time?

Aditya Madanapalle | aditya@digit.in

A

According to our current understanding, time as we know it began with the Big Bang. Physics does

not have a concept of before the Big Bang, because there is no way to measure such a thing. Yet, there are objects in the sky which appear to be older than the universe itself. Others have appeared soon after the Big Bang, when there was simply not enough material for their formation. There are some indications that exotic objects

such as some black holes could be remnants from a previous universe.

THE METHUSELAH STAR

One of the first objects thought to be older than the universe was a subgiant star moving towards the Earth. About 200 light years away, in the direction of the constellation of Libra is a star designated as HD 140283. What is unique about it is that it is one of the oldest stars known, and is extremely metal poor. In astronomy, the “metallicity” of a star is a shorthand for the presence of all elements heavier than hydrogen and helium. This is because most of the elements that we know

of were cooked in the hearts of stars, which then spread across the universe, into clouds of dust and nurseries of another generation of stars, which subsequently had more “metal” in them. The older stars can be considered to be polluted by the metals of a previous generation of stars. HD 140283 was dubbed the “Methuselah” star considering just how ancient it is.

The Methuselah star is so anemic, that it has only 1/250th the metal content as our Sun. This makes the star very, very old. Early estimates of the age of the star pegged it at 16 billion years old. The problem is, that the universe is only 13.787 billion years